

1. Consider the non-inverting feedback amplifier in Figure, where the actual OPAMP exhibits a finite output resistance R_{out} and a maximum output current capacity I_{max} . The feedback network consists of resistors $R_1 = 4 \text{ M}\Omega$ and $R_2 = 1 \text{ M}\Omega$, driving a total load capacitance of $C_L = 8 \text{ pF}$. The open-loop low-frequency gain of the op-amp is 80 dB, and its small-signal output resistance is $R_{out} = 50 \text{ k}\Omega$. The op-amp limits its maximum output current to $I_{max} = 160 \text{ }\mu\text{A}$.

(a) Assuming small-signal (linear) operation:

- Derive the expression for the closed-loop low-frequency voltage gain A_{CL} and calculate its numerical value.
- Calculate the small-signal time constant of the closed-loop step response.
- Estimate the small-signal output $V_{out}(t)$ for an input voltage step $V_{in}(t) = V_0 u(t)$ with amplitude $V_0 = 50 \text{ mV}$ after 1 ns.

(b) Calculate the maximum initial output slope $(dV_{out}/dt)|_{t=0+}$ during small-signal linear settling for $V_0 = 50 \text{ mV}$.

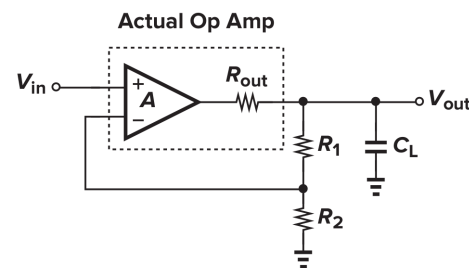
(c) Under large-signal operation:

- Determine the slew rate (SR) of the op-amp when driving the load capacitance C_L .
- Find the minimum input step amplitude $V_{0,crit}$ above which the output initial rate of change becomes limited by slewing rather than linear response.

(d) A large step input of amplitude $V_0 = 1.0 \text{ V}$ is applied at $t = 0$:

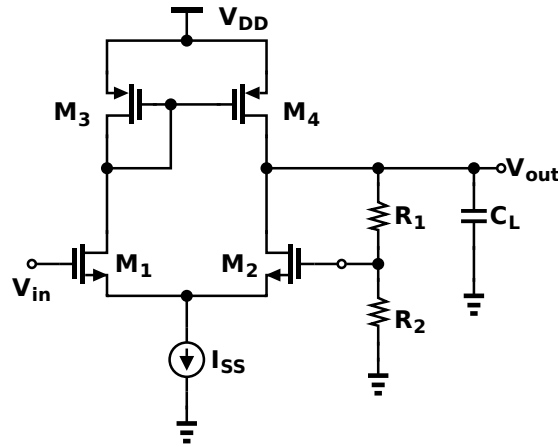
- Estimate the time interval t_{slew} during which the output ramps linearly.
- Sketch qualitatively the total step response $V_{out}(t)$, clearly identifying the slewing phase and the subsequent linear settling phase.

$$V_{out} \simeq V_0 \frac{A}{1 + A \frac{R_2}{R_1 + R_2}} \left[1 - \exp \left(- \frac{t}{\frac{C_L R_{out}}{1 + A R_2 / (R_1 + R_2)}} \right) \right]$$



2. Consider the non-inverting feedback amplifier configuration utilizing a simple CMOS op amp as shown in Fig. The circuit parameters are given as follows: Tail bias current $I_{SS} = 200 \mu\text{A}$, load capacitance $C_L = 5 \text{ pF}$, feedback resistors $R_1 = 3 \text{ M}\Omega$ and $R_2 = 1 \text{ M}\Omega$. The NMOS input differential pair transistors (M_1, M_2) have $\mu_n C_{ox}(W/L)_{1,2} = 4 \text{ mA/V}^2$. The PMOS current mirror load transistors (M_3, M_4) have early voltage parameters $V_A = |V_{AP}| = V_{AN} = 20 \text{ V}$.

- Calculate the positive slew rate (SR_+) and negative slew rate (SR_-) at the output node V_{out} .
- Determine the minimum differential input step voltage $\Delta V_{in,min}$ applied at V_{in} that fully turns off M_2 and forces the op amp entirely into the slewing regime.
- A positive input step of amplitude $V_0 = 1.2 \text{ V}$ is applied to V_{in} at $t = 0$. Calculate the slewing time duration t_{slew} required for node X to reach the point where M_2 turns back on ($V_{in} - V_X = \Delta V_{in,min}$).
- Once the circuit exits the slewing regime, it enters linear settling:
 - Calculate the small-signal open-loop output resistance under equilibrium bias conditions.
 - Derive the closed-loop time constant τ_{cl} during linear settling.
 - Determine the total time t_{total} taken from $t = 0$ for the output $V_{out}(t)$ to settle within 1% of its final steady-state value.



3. Consider the folded-cascode op amp depicted in Figure, driving a load capacitance $C_L = 4 \text{ pF}$. The NMOS input differential pair is biased with a tail current $I_{SS} = 300 \text{ }\mu\text{A}$, and each upper folding PMOS current source is biased to supply $I_P = 200 \text{ }\mu\text{A}$ under quiescent conditions. Assume all transistors have early voltage parameter $V_A = 25 \text{ V}$, and the input differential pair operates with an equilibrium overdrive voltage $V_{OV1,2} = 150 \text{ mV}$.

- Calculate the maximum positive slew rate (SR_+) at the output node V_{out} .
- Calculate the maximum negative slew rate (SR_-) at the output node V_{out} , and identify which specific bias current limits this rate.
- Determine the minimum condition on I_P relative to I_{SS} to guarantee symmetrical slewing ($SR_+ = SR_-$).
- Calculate the minimum differential input voltage step $\Delta V_{in,min}$ required to push the input stage completely into the slewing regime.

